

Best Practice for Distributed-Server Systems

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1. Introduction

The purpose of this document is to introduce best practices for installing Therefore™ on a distributed server system. A distributed-server system will typically include three layers, as well as primary and backup storage:

1. DMZ Layer

The DMZ Layer is located behind the infrastructure firewall and provides an access point between the Therefore™ system and the internet. External client software communicates with the DMZ server through http/https, while the DMZ server communicates with the Therefore™ Server using DCOM within the local network. The DMZ server runs the following Therefore™ components: Web Access, XML Web Service, MFP Manager, Mobile Manager and Conversion Service. To host Therefore™ Web Access, Microsoft IIS is required on this server. While using our Angular based web applications such as Therefore™ eForms or Therefore™ Dynamic Web View subdomain mode is only supported in a multi-tenant system. Subdomain mode is not supported in a single tenant system.



The Therefore™ services on the server in the DMZ layer could also be installed on different servers.

2. Application Layer

The application layer contains the Therefore™ Server itself. Internal client software communicates directly with the Therefore™ Server using DCOM. Internal http/https connections (such as mobile devices and MFP devices) are routed through the DMZ server which hosts the required services. The Therefore™ Server runs the following services: Therefore™ Server Service, Full-Text Indexing Service, and Conversion Service.

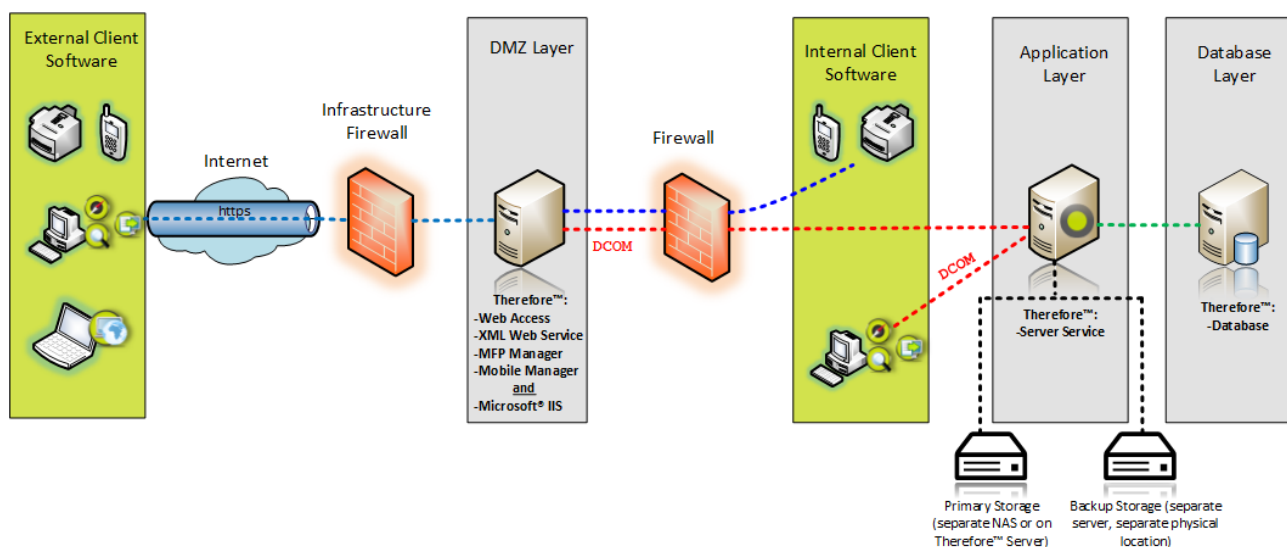
3. Database Layer

The database layer contains a server with the Therefore™ database. The Therefore™ Server connects to the database server.



The Therefore™ database can also be installed on the Therefore™ Server in the application layer. In this document, however, we are working under the assumption that the Therefore™ database is located on its own server.

The following graphic represents a typical Therefore™ installation on a distributed-server system.



4. Storage

Primary and backup storage should be configured according to the environment. In environments with fast network speeds, a NAS drive may be used for convenience and additional security. Otherwise, the primary storage can be configured on a separate drive on the Therefore™ Server. Backup storage should always be on a separate server, preferably in a different physical location. This is to prevent complete loss of data in case of hardware failure or destruction.

2. Installation

- On the Therefore™ DMZ system, install the latest version of Microsoft IIS .
- On the Therefore database system, install the desired database platform.
- Afterwards, run Windows Update on all servers and check for important updates.

2.1 Users and Groups

Therefore™ users and groups can either be local or part of a domain.

2.1.1 Groups

Therefore™ groups should be created for both administrators and users in order to easily define permissions within Therefore™ for different computer accounts.

- **Therefore™ Administrators:** Accounts in this group will be assigned administrative permissions within the Therefore™ system during installation.
- **Therefore™ Users:** Accounts in this group will be assigned user-level permissions within the Therefore™ system during installation.

2.1.2 Users

Ensure that the user accounts have strong, non-expiring passwords.

- **User1:** Used for DCOM communication between the two systems. This account has to be created on both systems with the same password (as long there is no domain account used).
Group memberships:
 - a. Member of Administrators group on the Therefore™ DMZ.
 - b. Member of Distributed COM Users group on the Therefore™ Server machine.
 - c. Member of Therefore™ Users group on the Therefore™ Server machine.
 - d. Member of Users group on the Therefore™ Server machine.
- **User2:** Has to be created on the Therefore™ Server machine.
Group memberships:
 - a. Member of Administrators group.
 - b. Member of Distributed COM Users group.
 - c. Member of Therefore™ Administrators group.

Permissions for SQL Server:

- a. Add User2 to the Security for database server
- b. Add User2 to the Database Login for the Therefore™ database (no Owned Schemas and Role Members)
- c. Right click on the Therefore™ database and open the Permissions tab. Give following permissions to User2:
 - i. Control



- The names User1 and User2 are representative; these should be changed to clear, self-explanatory names.

2.2 Therefore™ Server

1. Make sure the latest Windows updates have been installed.
2. Download the latest Therefore™ Server setup program.
3. Run Therefore™ Server setup. Make sure the following features are selected during installation:
 - i. Therefore™ Core: selected
 - ii. Therefore™ Server: selected
 - iii. Therefore™ Connect: none selected
 - iv. Therefore™ Adminstrate: Solution Designer, Console, Tenants Manager, Export Utility, Document Loader
 - v. Therefore™ Use: Navigator
 - vi. Therefore™ Integrate: none selected
 - vii. Languages: select all that is needed for Administering the system (via Tenants Manager)
 - viii. Documentation: select what is needed
4. After installation is finished, launch the Configuration Wizard.
 - i. At the license dialog enter the CustomerID and activate the license (in the case of re-installation, the license needs to be reset. For the license reset please contact the Therefore Support Team at support@therefore.net)
 - ii. Select the Database to use.
 - iii. Select the folders for Buffer, Cache, Preview and Full-Text and continue installation.
 - iv. Enable full-text search if needed.
 - v. Select the administrator and user groups defined previously.
 - vi. Change Therefore™ Server Service account/log on to User2 (either during setup or afterwards in services.msc and restart the Therefore™ Server Service).
 - vii. Deselect the default categories if not required.
 - viii. Transfer Folders are not recommended on hosted systems. These should not be configured.
 - ix. Check the box Enable Therefore™ Client Updates and enter the location of the folder where Therefore™ client updates will be stored. For more information regarding Therefore™ updates, refer to the Upgrade Manual.
 - x. Finish the configuration.
5. Make sure that the Therefore™ Full-Text Indexing Service and Therefore™ Conversion Service run under Local System account.



- If the Microsoft SQL Server is located on a different server, the Therefore™ Full-Text Indexing Service must be run by the User2 account.

2.3 Therefore™ DMZ

1. Make sure that the latest version of Microsoft IIS is installed.
2. Download the latest Therefore™ Server setup program.
3. Therefore™ Installation

- a. Make sure that the following features are selected during the installation
 - i. Therefore™ Core: selected
 - ii. Therefore™ Server: NOT selected (Important!)
 - iii. Therefore™ Connect: Web Access, XML Web, MFP Manager, Mobile Manager
 - iv. Therefore™ Administrate: none selected
 - v. Therefore™ Use: Navigator (for DCOM testing)
 - vi. Therefore™ Integrate: none selected
 - vii. Languages: select all that is needed
 - viii. Documentation: NOT selected
4. When installation is finished, don't choose "Launch Therefore™ Navigator".

2.4 Therefore™ Database

1. Make sure your selected database platform is installed.
2. Make sure the database server is accessible by the Therefore™ Server Service and the Therefore™ Full-Text Indexing Service by opening all necessary ports.

2.4.1 Configuration

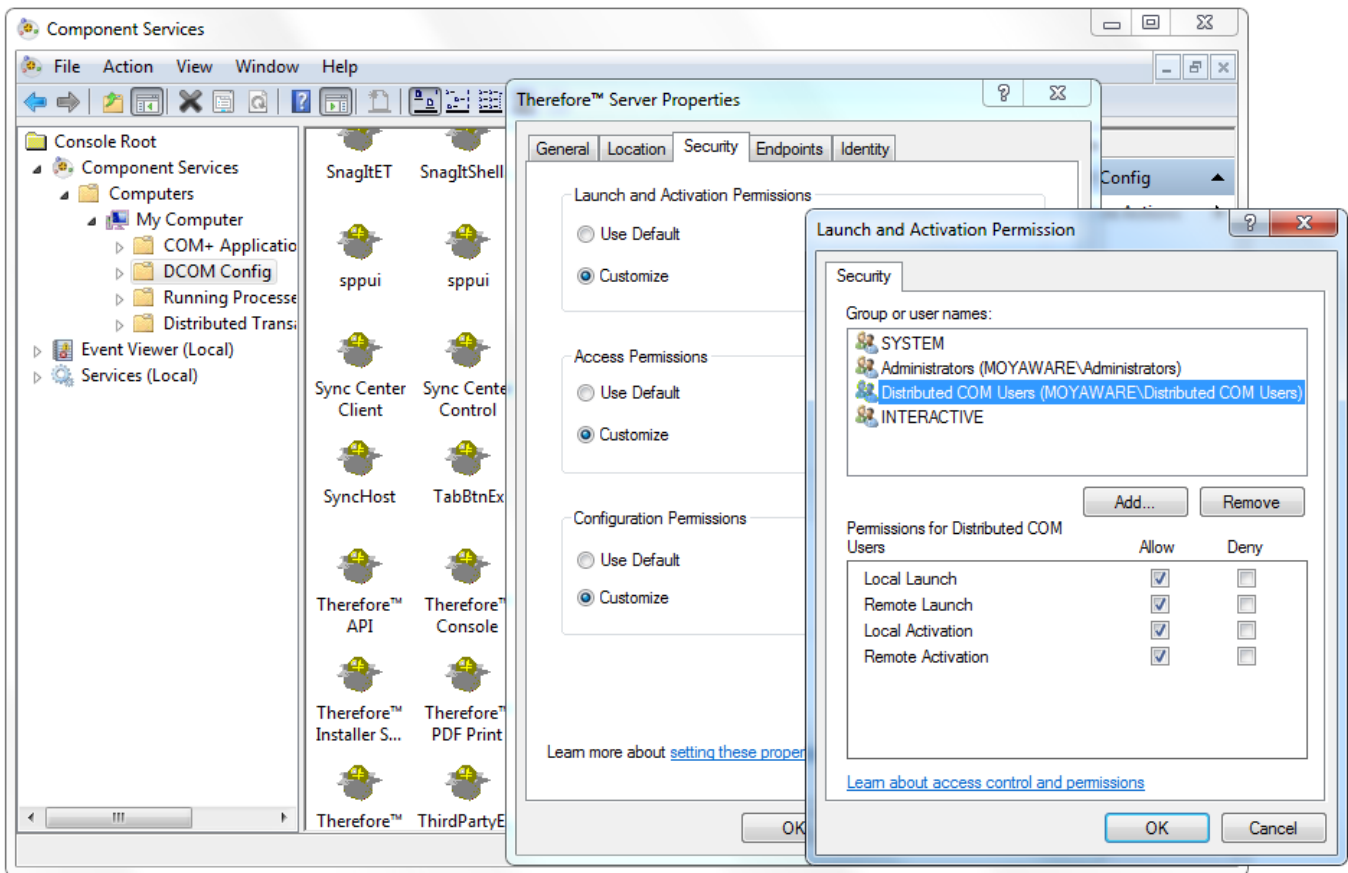
2.4.1.1 DCOM

It is necessary to configure a DCOM connection between the Therefore™ DMZ and the Therefore™ Server.

2.4.1.1.1 Therefore™ Server

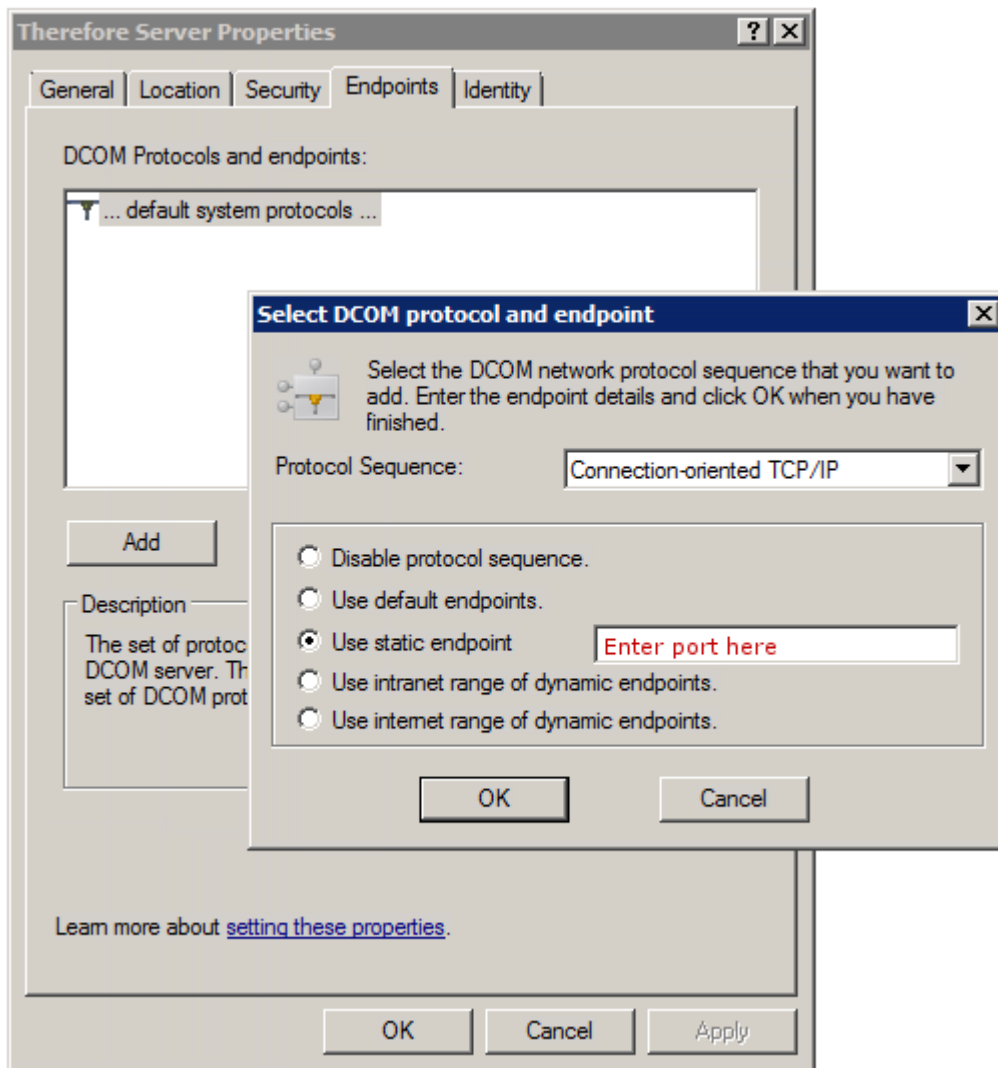
1. Configure DCOM Security
 - a. Go to Control Panel > Administrative Tools > Component Services or run the command `dcomcnfg`
 - b. Expand Component Services > Computers
 - c. Expand My Computer and select DCOM Config
 - d. Right-click on Therefore™ Server and select Properties
 - e. Click the Security tab
 - f. For Launch and Activation Permissions, select Customize, then Edit...
 - g. If Distributed COM Users is not in the list of group or user names then click the Add... button
 - i. Make sure that the location is set to the local computer (if asked for a network password click Cancel)
 - ii. Enter "Distributed COM Users" in the Enter the object names to select box, select Check Names and then OK
 - h. Give the following permissions to the Distributed COM Users group: Local Launch, Remote Launch, Local Activation and Remote Activation. Click OK when done
 - i. For Access Permissions, select Customize, then Edit...
 - j. If Distributed COM Users is not in the list of group or user names then click the Add... button
 - i. Make sure that the location is set to the local computer (if asked for a network password click Cancel)
 - ii. Enter "Distributed COM Users" in the Enter the object names to select box, select Check Names and then OK

- k. Give the following permissions to the Distributed COM Users group: Local Access and Remote Access. Click OK when done



2. Configure DCOM protocol and endpoint

- a. If the architecture requires static endpoints used by DCOM you have to do the following steps:
 - i. Go to Control Panel > Administrative Tools > Component Services
 - ii. Expand Computers > My Computer > DCOM Config
 - iii. In the right pane search for Therefore Server
 - iv. Right-click on it and select Properties
 - v. Go to the Endpoints tab
 - vi. Click Add
 - vii. Select Connection-oriented TCP/IP
 - viii. Select Use static endpoint
 - ix. Enter the port you have opened in your firewall for DCOM in the box next to the selected option



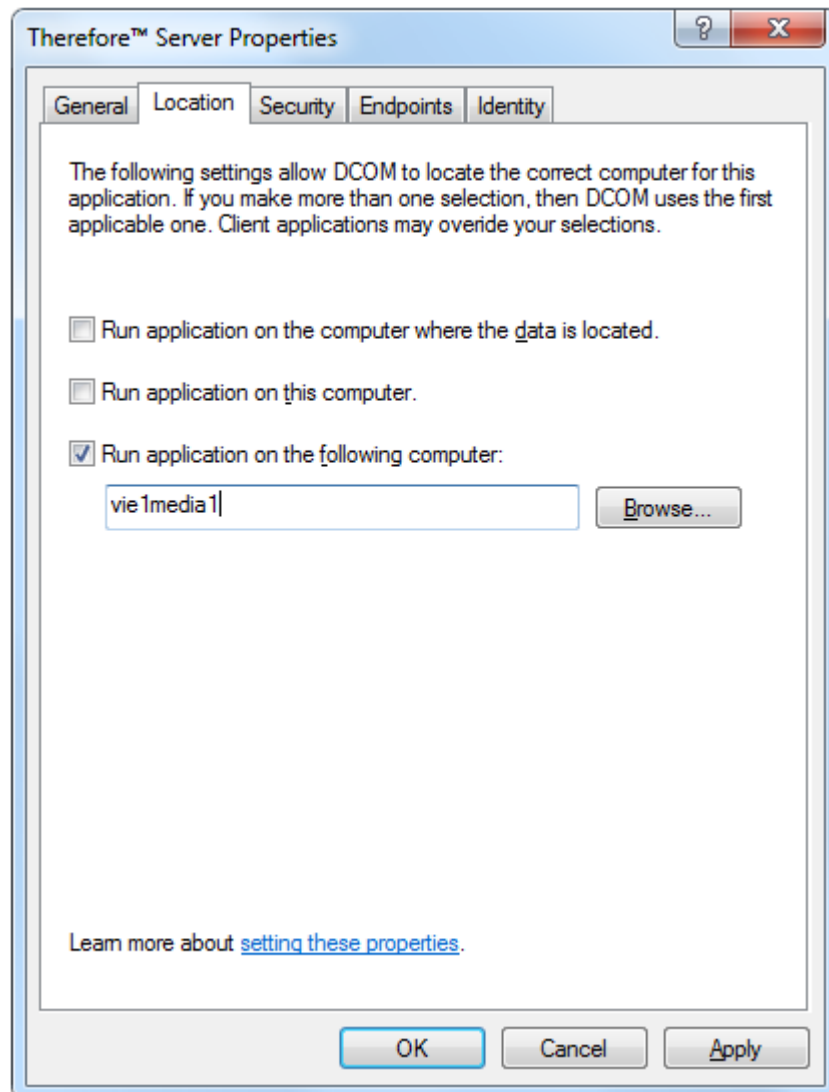
- x. Click OK
- xi. Click OK



It is also necessary to open port 135 in the firewall.

2.4.1.1.2 Therefore™ DMZ

- Open Component Services (run dcomcnfg.exe).
- Expand Component Services, expand Computers, expand My Computer, click on DCOM Config.
- Find Therefore™ Server, right-click Properties.
- Under Location tab tick Run application on the following computer:, and type in the name of the App Server. Make sure only this box is ticked. Save the settings.



2.4.1.1.3 Check DCOM Connection

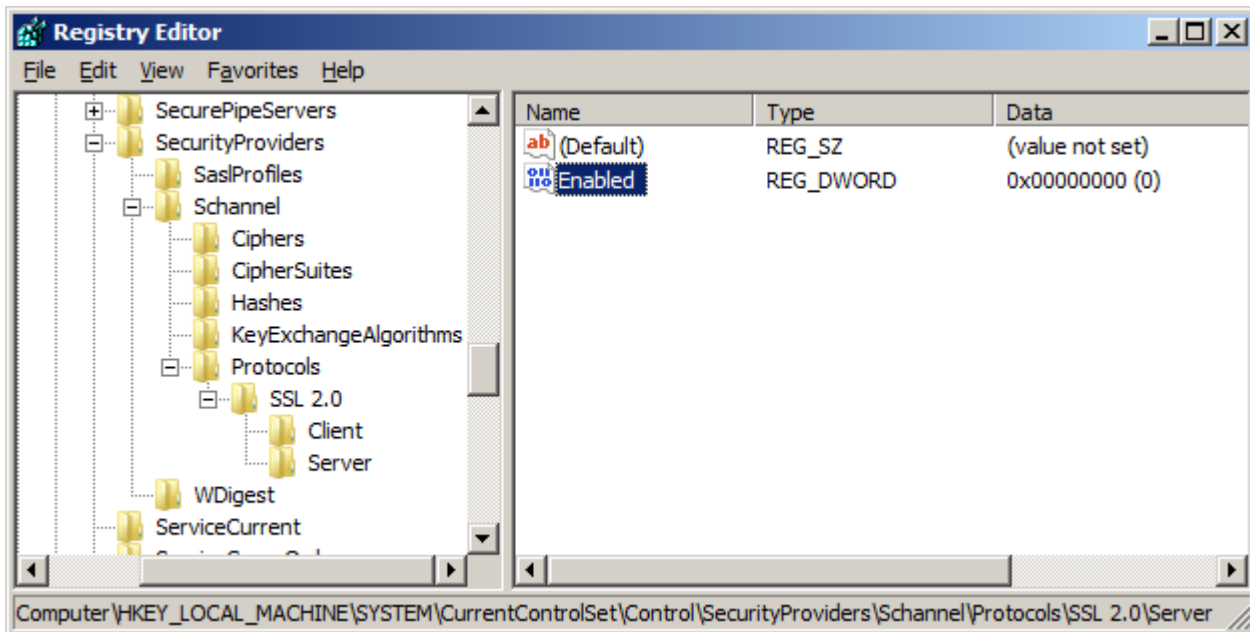
1. Run the Therefore™ Navigator as User1 and check that the DCOM connection is working.
2. Then set up the Therefore™ connection settings if necessary (IP address of the server, and Tenant Name if Multi-Tenant mode has already been activated).
3. Use the User1 as the log on for XML Web Service, Web Service and MFP Manager.
4. Make sure that HKLM\Software\Therefore\Client in the Windows® registry includes the same connection settings as HKLM\Software Wow6432Node\Therefore\Client

2.4.1.2 Disable SSL v2.0 (Therefore™ DMZ)



This action is performed on the Therefore™ DMZ server.

1. Click Start, click Run, type regedit, and then click OK.
2. Locate the following registry key/folder:
[HKEY_LOCAL_MACHINE\System\CurrentControlSet\Control\SecurityProviders\SCHANNEL\Protocols\SSL 2.0](#)
3. Right-click on the SSL 2.0 folder and select New and then click Key. Name the new folder Server.
4. Inside the Server folder, click the Edit menu, select New, and click DWORD (32-bit) Value.
5. Enter Enabled as the name and hit Enter.
6. Ensure that it shows 0x00000000 (0) under the Data column (set by default). If it doesn't, right-click, select Modify and enter 0 as the Value data.



7. Restart the computer.

2.4.1.3 Secure Connection Configuration (Therefore™ DMZ)



The actions in sections 3.3.1 to 3.3.6 are performed on the Therefore™ DMZ server.

2.4.1.3.1 Configuration of HTTPS on Microsoft IIS

1. Install the server certificate according to the instructions of your certificate provider.
2. Go to Default Web Site (or the desired site if it is a different one)
 - a. Right-click on it and from the context menu select Edit Bindings...
 - b. Add a new binding by clicking the Add... button
 - c. For the type select https
 - d. Make sure the IP address is set to All Unassigned
 - e. Set the port to 443
 - f. For the SSL Certificate select the certificate which should be used for this site
 - g. Do not remove http binding
 - h. Close the dialog once done
3. Subdomain Mode configuration

If required, <http://demo.yoursite.com> can be used instead of <http://yoursite.com/TWA> for easier web client access

- a. Enable Subdomain Mode
 - i. Edit web.config file that can be found under C:\inetpub\wwwroot\TWA
<add key="SubdomainMode" value="true">
- b. Go to IIS Home
- c. Select Default Web Site and from the Actions pane choose Basic Settings
 - i. Change Application Pool to Therefore™
 - ii. Change Physical Path to C:\inetpub\wwwroot\TWA

This will remove the TWA from the resulting URL

- d. Convert Client and MFP and Portal directories to applications by selecting the Convert to Application from the context menu
 - i. Make sure that Application Pool is set to Therefore™
- e. Remove the TWA application by selecting Remove from the context menu



While using our Angular based web applications such as Therefore™ eForms or Therefore™ Dynamic Web View subdomain mode is only supported in a multi-tenant system. Subdomain mode is not supported in a single tenant system.

2.4.1.3.2 Certificate Thumbprint Retrieval

1. Snap-in Certificates
 - a. Select Start > Run.. and type MMC
 - b. Choose File > Add/Remove Snap-in...
 - c. From the available snap-ins select Certificates and choose Add>
 - d. Select Computer account and then Local computer
2. Retrieve the Certificate Thumbprint
 - a. Expand the Certificates > Personal > Certificates and double-click on the certificate
 - b. Go to the Details tab
 - c. Scroll down to Thumbprint
 - d. Copy the Thumbprint from the box underneath

Make sure to remove all white space characters before performing the following steps.

2.4.1.3.3 Configuration of HTTPS - XML Server

1. Browse to C:\Program Files\Therefore directory and edit TheXMLServer.exe.config file. The highlighted lines below in **yellow** simply need to be deleted and then the file re-saved. If you want to use an XML Transfer Folder, the lines highlighted in **green** also need to be deleted. For more information about using transfer folders, please see the Installation Guide.



- "TheXMLServer.exe.config" file is edited on a server.
- This step is only required if multi-tenancy will be used. See the Multi-Tenancy manual for more details.

```
<?xml version="1.0"?>
<configuration>
  <appSettings>
    <!-- Uncomment in a multi-tenant environment -->
    <!--
    <add key="ServiceName" value="TheXMLServer"/>
    <add key="Port" value="8001"/>
    <add key="UseHTTPS" value="0"/>
    -->
    <!-- Uncomment the following if using XML Transfer Folder -->
    <!--
    <add key="TransferFolder" value="C:\Therefore\TransferFolder"/>
    <add key="UploadFolderCleanupTime" value="24"/>
    <add key="DownloadFolderCleanupTime" value="24"/>
    -->
  </appSettings>
```

- a. Set the Port to 443
- b. Set the UseHTTPS value to 1

2.4.1.3.4 Configuration of HTTPS - MFP Manager

1. Browse to C:\Program Files\Therefore directory and edit TheMFPManager.exe.config file
 - a. Set UseStandardXMLPort value to 1
 - b. Set FilePort value to 8091
 - c. Set UseHTTPS value to 1
 - d. Set CertStore value to My
 - e. Set CertLoc value to LocalMachine
 - f. Set CertPrint value to the Thumbprint of the certificate
 - g. Leave DefaultDomain commented or remove it if not required

2.4.1.3.5 Configuration of HTTPS - Mobile Manager

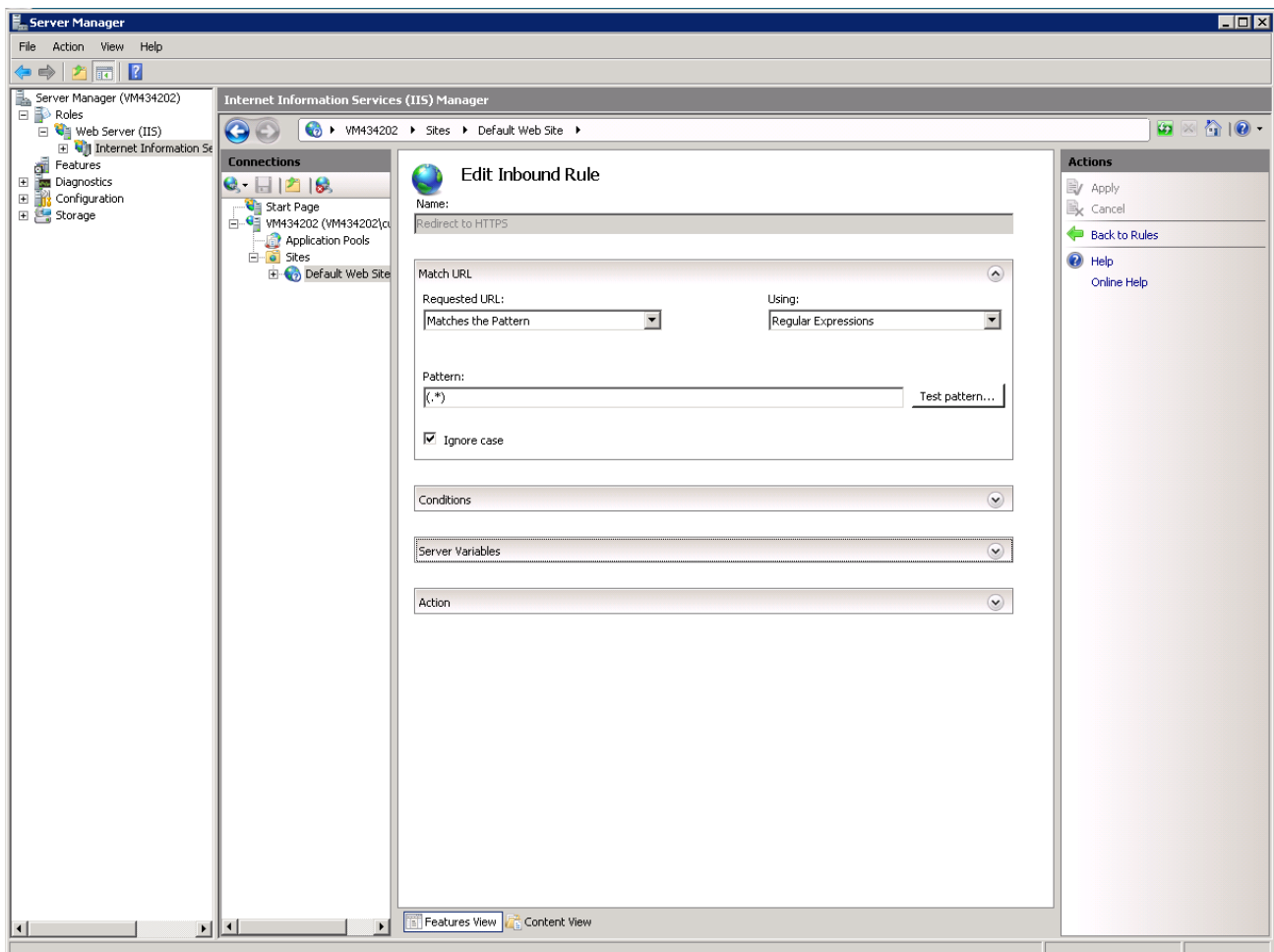
1. Browse to C:\Program Files\Therefore directory and edit TheMobileManager.exe.config file
 - a. Uncomment the multi-tenant environment part



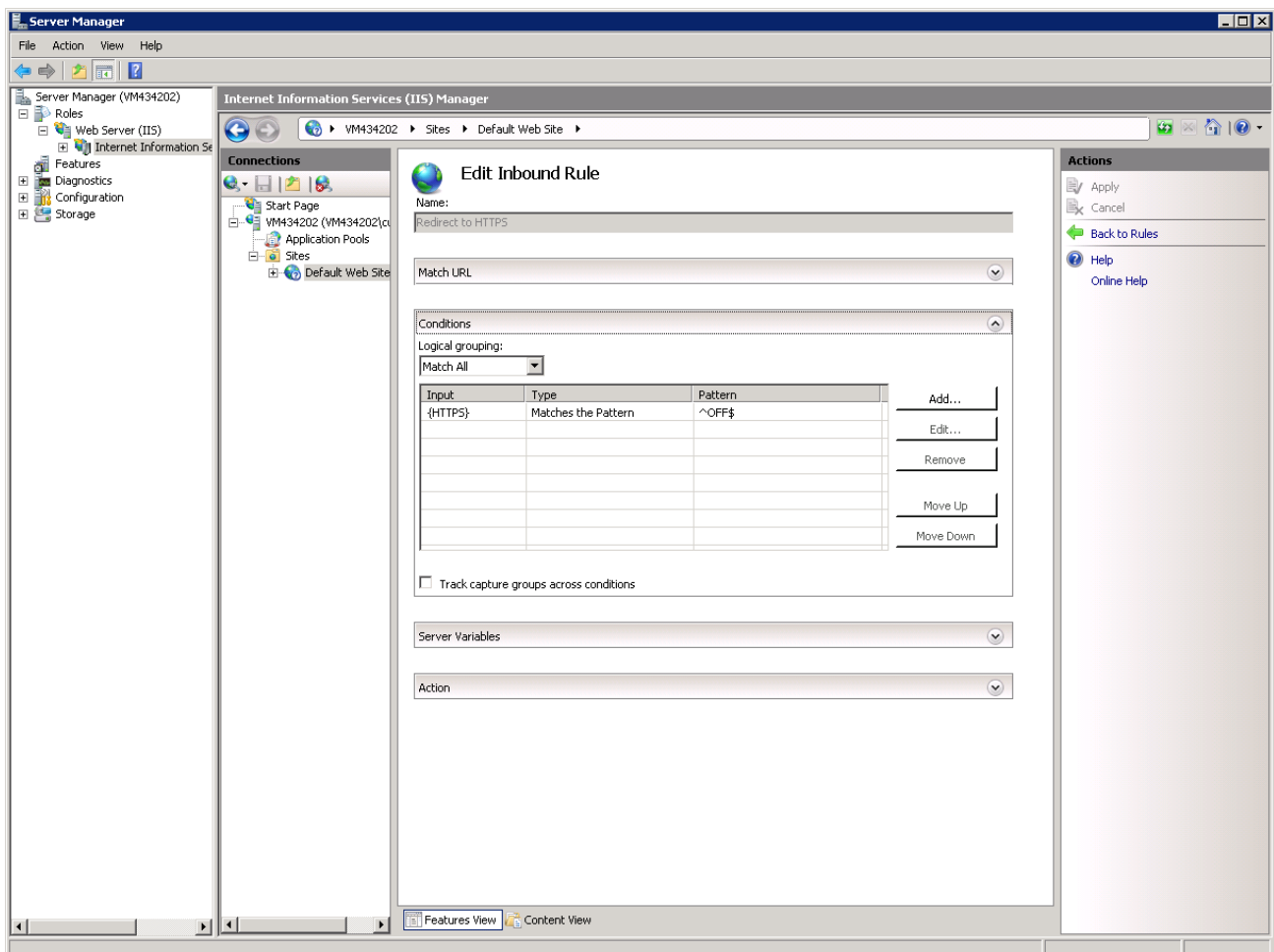
This step is only required if multi-tenancy will be used.

2.4.1.3.6 Set up HTTP to HTTPS Redirection

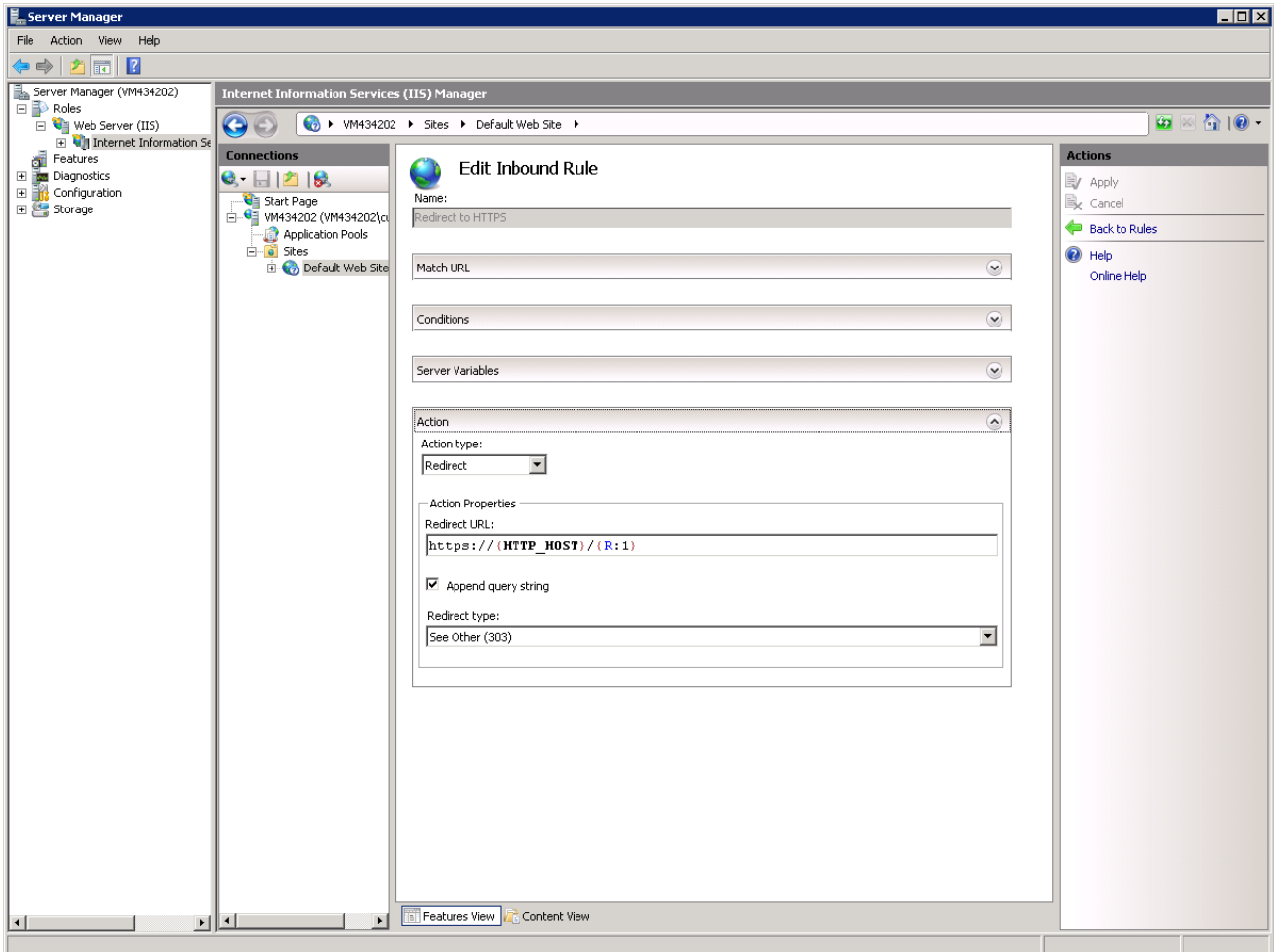
1. Download and install "URL Rewrite" from Microsoft (<http://www.iis.net/download/URLRewrite>)
2. In the Feature View on Microsoft IIS for the Default Web Site, open the URL Rewrite
 - a. Click Add Rule(s)...
 - b. Choose Blank Rule and press OK
 - c. Enter the name: Redirect to HTTPS
 - d. In the Match URL section:
 - i. Make sure Matches the Pattern is selected
 - ii. Make sure Using: is set to Regular Expressions
 - iii. At Pattern: enter (.*)



- e. In the Conditions section, click Add... and then:
 - i. Condition input has to be {HTTPS}
 - ii. Type needs to be set to Matches the Pattern
 - iii. Pattern has to be ^OFF\$
 - iv. Once you are done Click OK



- f. In the Action section:
 - i. For the action type choose Redirect
 - ii. Redirect URL has to be set to `https://{HTTP_HOST}/{R:1}`
 - iii. For redirect type choose See Other (303)



- 3. Apply the rule
- 4. Make sure IIS SSL Settings are NOT set to Require SSL

2.4.1.4 Mail Server Configuration

If email notifications are required, a mail server needs to be configured.

2.4.1.5 Security Checks

The following security checks are recommended:

- 1. Use Microsoft Baseline Security Analyzer
- 2. Use Microsoft Best Practices Analyzer
- 3. Read the Therefore™ Security white paper

2.4.1.6 Application Server Configuration

2.4.1.6.1 Create Admin User

1. Open the Therefore™ Solution Designer.
2. Expand Design and then Users and Groups.
3. Right-click on Users and from the context menu select Add User...
4. Enter a name and a password for a user e.g.

Name: firstname.lastname

Password: Generate a random password by using the
<http://www.pctools.com/guides/password/>

5. Select the checkbox User must change the password at next logon.

2.4.1.6.2 Assign Permissions to Admin User

1. Right-click on the root node Therefore and from the context menu select Security...
2. Click the Add... button and browse for the newly created user.
3. Give full permissions to this user:

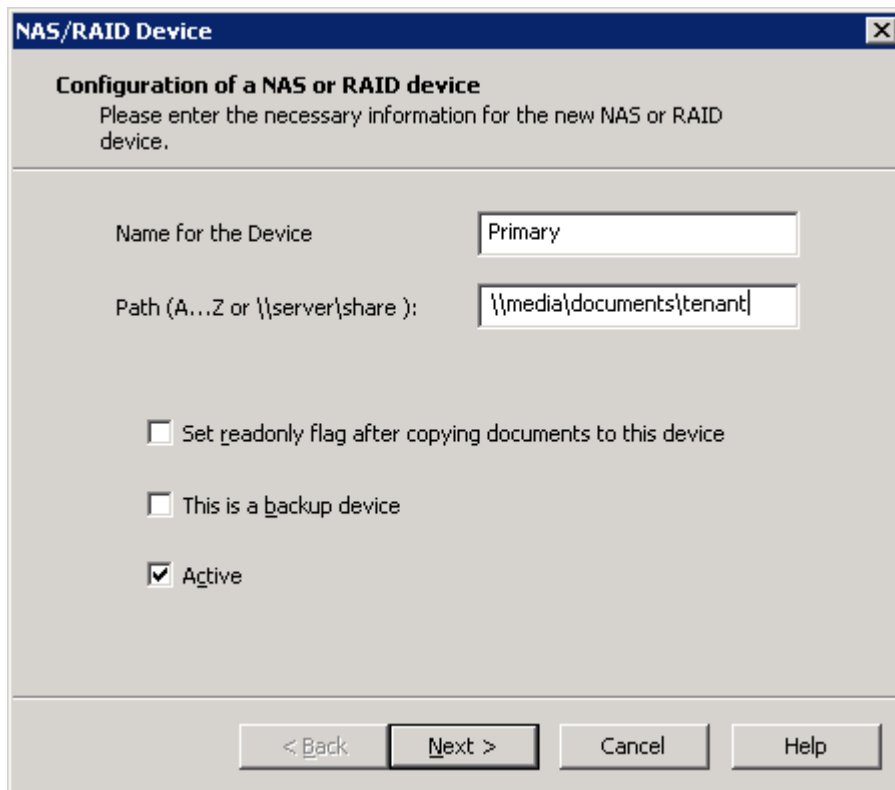
Permission	Allow
User/Read	Yes
User/Write	Yes
Operator	Yes
Administrator	Yes
Set permissions	Yes

4. Click OK to save the configuration.
5. Expand the Storage node and then select Devices.
6. Right-click on the Devices node and from the context menu select Security...
7. Click the Add... button and browse for the newly created user.
8. Set all permissions to Deny except User/Connect permission:

Permission	Allow	Deny
User/Connect	Yes	
Operator		Yes
Administrator		Yes
Set permissions		Yes

2.4.1.6.3 Set up Migration

1. Create the subdirectory on the final media where documents should be stored e.g. NAS drive
2. Create a new Primary storage
 - a. Under the Storage node right-click on Devices and from the context menu select New Device...
Name for the device: Primary
Path: \\<path to the final media>\
Make sure that only Active is selected



- b. Click Next
 - c. Add a new storage area e.g. Area1
 - d. Click Finish
3. Create a new Backup storage
 - e. Use the same procedure as for the Primary storage creation, but make sure that This is a backup device checkbox is selected



We recommend creating a backup device in a different data center or security zone.

4. Create a new Storage Policy
 - f. Under the Storage node right-click on Storage Policies and from the context menu select New Storage Policy...

Storage Policy Name: give a name to the policy e.g. Primary documents migration

Primary Storage: select previously created storage area e.g. Primary.Area1

Backup Storage: select previously created storage area e.g. Backup.Area1

g. Click OK

5. Drag all categories from <Keep in Buffer> to the newly created storage policy

Check the storage policy ID: usually it should be 1

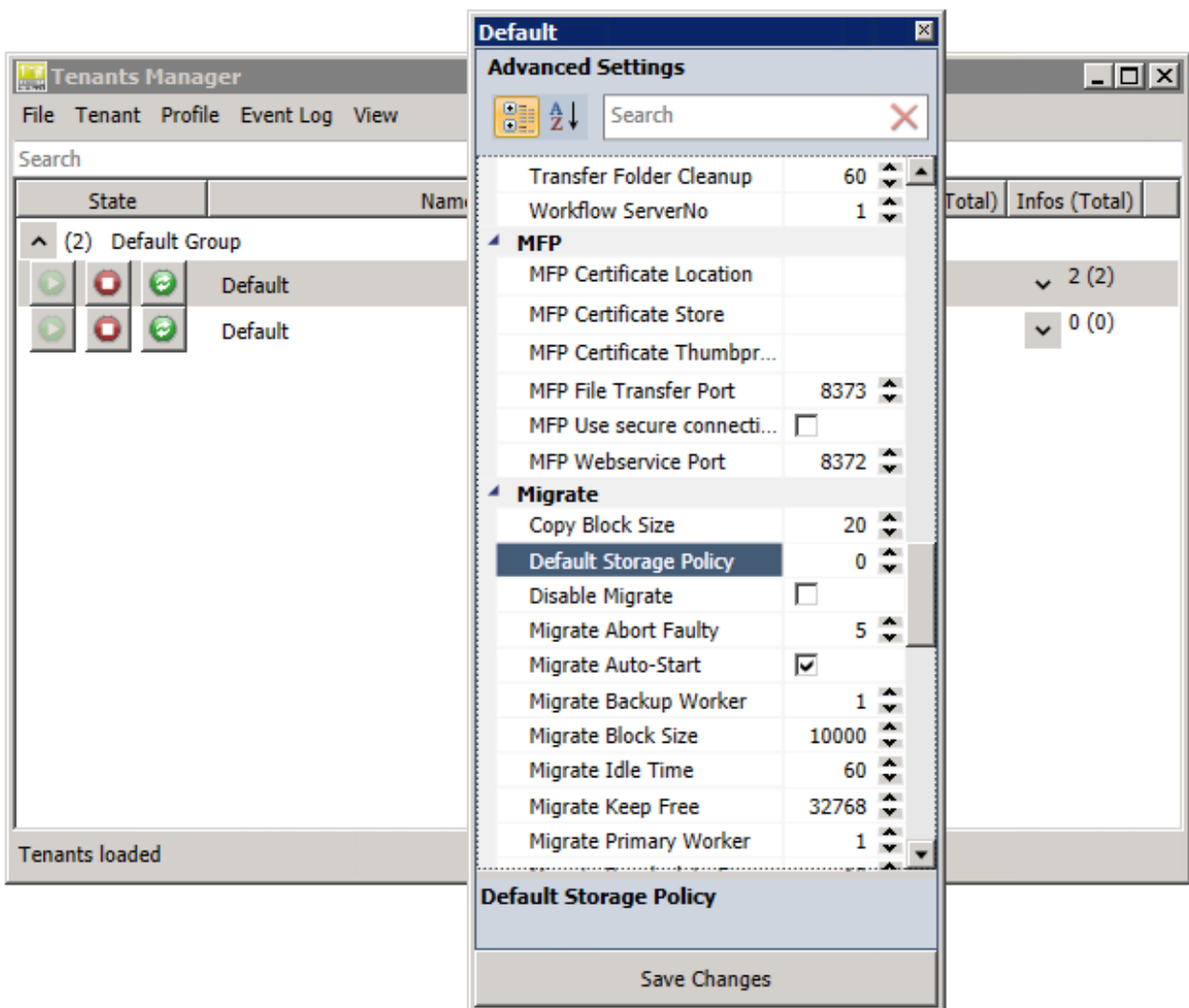


Steps 6 to 9 are required only if multi-tenancy will be used.

6. Open the Tenants Manager

7. Select the tenant and from the menu select View and then Advanced Settings

8. Under Migrate section set the Default Storage Policy to the ID of the previously created storage policy



9. Click on Save Changes

2.4.1.6.4 System Check

1. Log in via the Therefore™ Web Access and save a document.
2. Log in via Therefore™ Navigator and retrieve the previously saved document.
3. Test the MFP connection to the DMZ Server. Use port 8091.

3. Final Checklist

Before going live perform the following checks:

- Provision a new tenant (only if using multi-tenancy).
 - Therefore™ Solution Designer
 - o Check the connection via HTTP(S)
 - o Create new groups/users
 - o Design categories/workflows
 - Paper documents
 - o Test scanning via the Therefore™ Capture Client
 - o Test MFP integration
 - Electronic documents
 - o Test Windows Explorer integration
 - o Test Microsoft Office integration
 - Retrieve and view documents
 - o Via the Therefore™ Navigator
 - o Via the MFP device
 - o Via the Therefore™ Web Access
-  Test cross-category search, full-text search and index search.
- Workflow
 - o Approve an invoice via the Therefore™ Navigator, Therefore™ Web Access, and mobile device
 - o Test e-mail notifications
 - Migration
 - o Run migration and retrieve the document from the final media
 - Redirection
 - o Make sure the http to https redirection is still working
 - Change user password.